Mechanics Year 1 · Chapter 9

Constant Acceleration

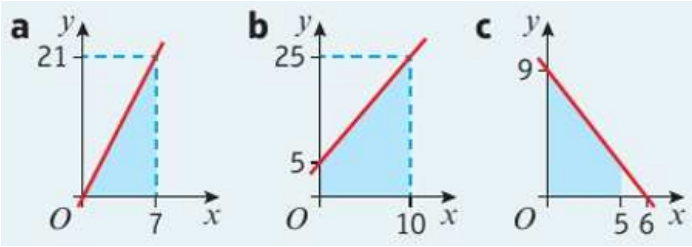
Displacement–time and velocity–time graphs

The constant-acceleration equations

Vertical motion under gravity

Starter 2

1. For each graph, find its gradient and the shaded area.



1

2. A car travels for 45 minutes at an average speed of 35 mph. Find the distance.

2

3. Solve $3x - 2y = 9, x + 4y + 4 = 0$.

3

4. Solve $2x^2 + 3x - 7 = 0$ to 3 significant figures.

4

RECAP: Displacement-Time Graphs 3

Displacement–time graphs

Describe each graph: a horizontal line; a rising straight line; a rising curve whose gradient increases.

Motion

Velocity is the gradient

$$v = \frac{ds}{dt}$$

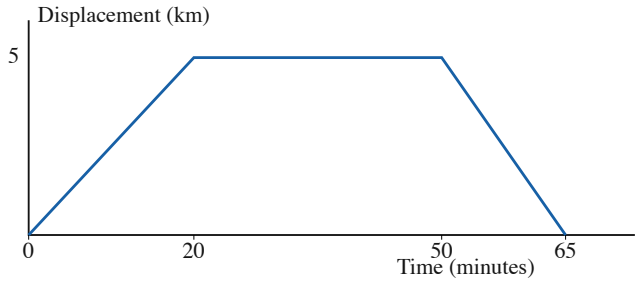
$$\text{Average velocity} = \frac{\text{displacement}}{\text{time}}$$

$$\text{Average speed} = \frac{\text{total distance}}{\text{time}}$$

Returning to the starting point gives zero overall displacement, even when the distance travelled is not zero.

Example 4

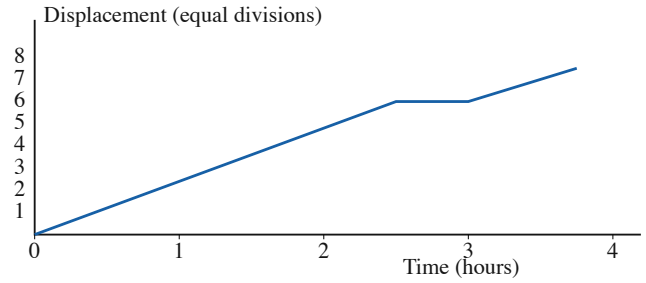
A cyclist travels 5 km in a straight line in 20 minutes, waits for half an hour, then returns in 15 minutes. Find each stage's average velocity, the whole journey's average velocity, and its average speed.



Stages

Whole journey

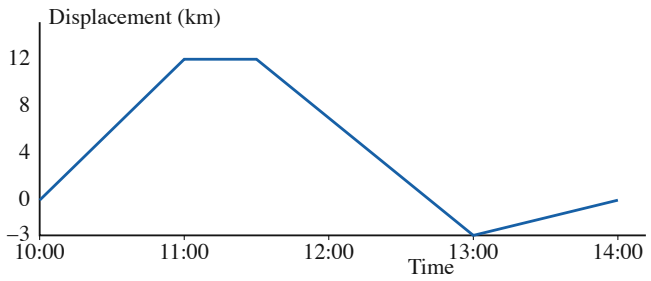
2. Khalid drives from home towards a hotel for 2.5 hours at an average velocity of 60 km/h, stops for lunch, then continues. The graph uses equal vertical divisions.
   a. Find the hotel's displacement from home.
   b. Find the average velocity for the journey.



a

b

3. Sarah cycles north from home at 10:00. Use the displacement–time graph.
   a. Find her velocity from 10:00 to 11:00.
   b. Estimate when she passes home on the return.
   c. Find the last two stage velocities.
   d. Find her average speed overall.

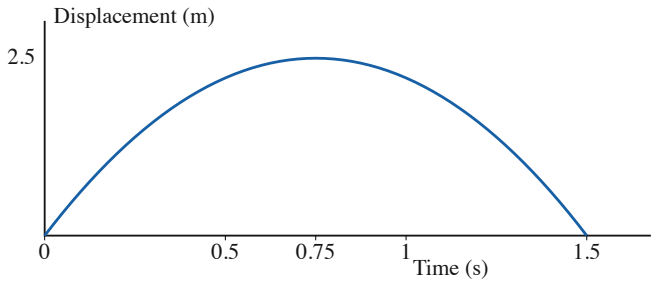


a

b

c and d

4. A ball is thrown vertically upwards and returns to the ground. Use its displacement–time graph.
   a. Find the maximum height and its time.
   b. State the velocity at the highest point.
   c. Describe the motion before and after the highest point.



a and b

c

RECAP: Velocity-Time Graphs 8

Velocity–time graphs

A line on the time axis means stationary. A horizontal line above the axis means constant positive velocity. A rising straight line means constant positive acceleration.

Gradient and area

$$a = \frac{dv}{dt}$$

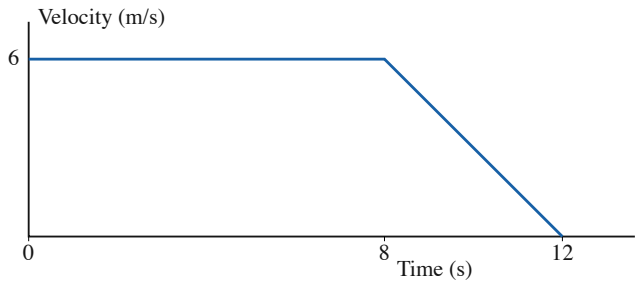
Signed area under a velocity–time graph gives displacement.

To find total distance, add the magnitudes of areas above and below the time axis.

Area under a speed–time graph gives distance. These coincide only when the velocity does not change sign.

Examples 9

A cyclist travels at 6 m/s for 8 s, then decelerates uniformly to rest over 4 s. Find her displacement and deceleration.



Displacement

Deceleration

Algebraic Example 10

A particle accelerates uniformly from rest to 8 m/s in T s, stays at 8 m/s for $5T$ s, then decelerates uniformly to rest over 40 s.

a. Sketch the velocity–time graph.

b. Given total displacement 600 m, find T.

Graph

a

b

Test Your Understanding 11

A car travels 2145 m in 120 s. It accelerates uniformly from rest for 30 s to 22 m/s, maintains this for T s, then decelerates uniformly to rest.

a. Sketch its speed–time graph.

b. Find T.

A motorcycle starts 10 s later from the same point, accelerating uniformly from rest. It passes the car 990 m from the start while the car is travelling at 22 m/s.

c. Find the motorcycle's journey time to that point.

d. Find its acceleration.

Graph

a and b

c

d

Exercise 9B – Y1 Mech Pages 135-136 12

A car travels at 20 m/s for 25 s, decelerates at $0.4\ \text{m/s}^2$ to 8 m/s, stays at 8 m/s for 60 s, then accelerates uniformly to 20 m/s. The total distance is 1960 m.

a. Sketch the speed–time graph.

b. Find the deceleration time.

c. Find the total journey time.

Graph

a and b

c

3. A cyclist accelerates uniformly from rest to 8 m/s in 20 s, maintains 8 m/s for 40 s, then decelerates uniformly to rest in 15 s. Find:
   a. the initial acceleration;
   b. the final deceleration;
   c. the displacement after 75 s.

a and b

c

4. A motorcyclist accelerates uniformly from rest to 30 m/s in 15 s, maintains that velocity for T s, then decelerates uniformly to rest in 25 s.
   a. Sketch the velocity–time graph.
   b. If the total distance is 2.4 km, find the total time.

Graph

a

b

5. A train starts from rest and accelerates at $0.6\ \text{m/s}^2$ for 20 s. It maintains its velocity for T s, then decelerates uniformly to rest over 40 s. Stations are 4.2 km apart.
   a. Sketch the velocity–time graph.
   b. Find T.
   c. Find the distance at constant velocity.

Graph

a

b and c

6. A particle accelerates uniformly from rest to 10 m/s in 15 s. It maintains that velocity, then decelerates uniformly to rest. The constant-velocity duration is four times the deceleration duration.
   a. Sketch the velocity–time graph.
   b. Given total displacement 480 m, find the total time.

Graph

a

b

7. A particle travels 100 m. It accelerates uniformly from u m/s to 10 m/s in 3 s, maintains 10 m/s for 7 s, then decelerates uniformly to rest in 2 s. Find:
   a. u;
   b. its acceleration during the first 3 s.

a

b

8. A motorcyclist and car start from rest at the same junction at the same time. The motorcyclist accelerates at $3\ \text{m/s}^2$ for 8 s, then keeps a constant velocity. The car accelerates uniformly to 30 m/s in 20 s, then maintains this velocity.
   a. Sketch both velocity–time graphs.
   b. The car passes the motorcyclist by a pedestrian. Find the pedestrian's distance from the junction.

Graph

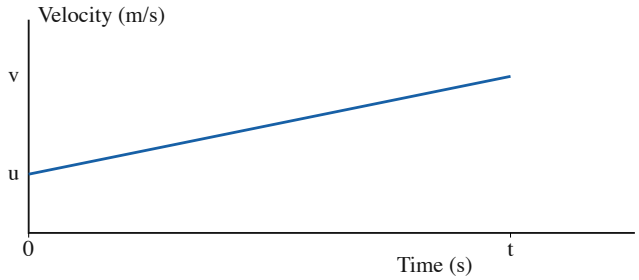
a

b

Constant-acceleration equations

s: displacement; u: initial velocity; v: final velocity; a: constant acceleration; t: duration.

Use a consistent positive direction and the same time interval for every quantity.



Derive two equations using the straight velocity–time graph.

Derivation

Examples 20

A cyclist accelerates uniformly from 4 m/s to 7.5 m/s in 40 s. Find:

a. the distance travelled;

b. the acceleration.

a

b

Examples 21

A particle passes A at 8 m/s and decelerates at $1.5\ \text{m/s}^2$. It reaches B at 2 m/s. Find:

a. the time from A to B;

b. the distance AB.

It continues with the same acceleration and reaches C 6 s after passing A. Find:

c. its velocity at C;

d. its displacement from A and total distance travelled.

a and b

c and d

Test Your Understanding 22

A car starts from rest at traffic lights and accelerates uniformly. After 30 s it passes a speed trap at 45 km/h. Find:

a. its acceleration;

b. the distance to the speed trap.

Units

a and b

9. A skier travels straight up a hill at 16 m/s and decelerates uniformly to rest over 40 s. Find:
   a. the deceleration;
   b. the distance travelled.

a

b

10. A particle moves with constant acceleration through A, B, C. Its velocities are 2, 7 and 11 m/s respectively. It takes 20 s from A to B. Find:
    a. the acceleration;
    b. the time from B to C;
    c. the distance AC.

a

b and c

11. A particle travels from A to B with acceleration $1.5\ \text{m/s}^2$ for 12 s, then from B to C with a different constant acceleration for 10 s. Velocities at A, C are 1 and 43 m/s. Find:
    a. the velocity at B;
    b. the acceleration from B to C;
    c. the distance AC.

a and b

c

12. A cyclist accelerates uniformly from rest to 5 m/s over 20 s at $x\ \text{m/s}^2$. She then decelerates uniformly to rest at $\frac{1}{2}x\ \text{m/s}^2$. Find:
    a. x;
    b. the total distance.

a

b

13. A particle passes A, B, C at 20, 30 and 45 m/s with constant acceleration. The times from A to B and B to C are t_1, t_2.
    a. Show that $t_1/t_2 = 2/3$.
    b. Given $t_1 + t_2 = 50$ s, find AB.

a

b

"suvat" equations (Part 2) 28

Eliminate time

$$s = \frac{u+v}{2}\frac{v-u}{a}$$

No time

Eliminate final velocity

$$s = \frac{u+(u+at)}{2}t$$

No final velocity

Eliminate initial velocity

$$s = \frac{(v-at)+v}{2}t$$

No initial velocity

Examples 29

1. A particle accelerates at $5\ \text{m/s}^2$ from 3 to 18 m/s. Find the distance.

1

2. A particle passes O at 13 m/s towards A, where $OA = 20$ m, with acceleration $-4\ \text{m/s}^2$. Find the times at A and the return time to O.

2

Test Your Understanding 30

A lorry passes A at u m/s and B at 34 m/s, 10 s later, with constant acceleration. Given $AB = 240$ m, find:

a. u;

b. the time from A to the midpoint of AB.

a

b

11. A particle passes the origin at 12 m/s with acceleration $-5\ \text{m/s}^2$. Find the times at $x = 8$ m and the velocity at $x = -8$ m.

11

12. Another particle passes the origin at 14 m/s with acceleration $-4\ \text{m/s}^2$. Find the difference between its times at $x = 22.5$ m and the total distance travelled between them.

12

13. A car passes consecutive points A, B, C with constant acceleration. $AB = 100$ m, $BC = 300$ m. Its speeds at B, C are 14 and 20 m/s. Find:
    a. the acceleration;
    b. the time from A to C.

a

b

14. Particles P, Q have accelerations 2 and $3.6\ \text{m/s}^2$. At $t = 0$, P passes A at 4 m/s. One second later Q passes A at 3 m/s in the same direction.
    a. Express each displacement in terms of t.
    b. Find when they meet.
    c. Find the distance of the meeting point from A.

a

b and c

15. An orienteer passes checkpoints P, Q, R in a straight line with constant acceleration. $PQ = 2.4$ km and $QR = 11.5$ km. The corresponding times are 1 hour and 1.5 hours. Find her acceleration and speed at P.

Set up

Solve

Vertical motion under gravity

Model the body as a particle and neglect air resistance. Near Earth, use $g = 9.8\ \text{m/s}^2$, unless the question specifies another value.

Acceleration is downward throughout the flight, including at the highest point.

Choose a positive direction

Up positive: $a = -g$. Down positive: $a = +g$.

At the highest point, $v = 0$; the acceleration is still downward.

Keep unrounded values during calculation and round final answers as the question requests.

The constant-acceleration equations apply separately to each uninterrupted stage.

Example 36

A book falls from rest from a shelf 1.4 m above a floor. Find:

a. the time to reach the floor;

b. its impact speed. Neglect air resistance.

a

b

Further Example 37

A ball is projected upwards at 21 m/s from a point 7 m above the ground. Neglect air resistance. Find:

a. the greatest height above the ground;
b. the time of flight.

a

b

A further common type of question. . . 38

A ball is projected vertically upwards from ground level at 20 m/s. Find how long it is at least 10 m above the ground. Neglect air resistance.

Boundary times

Duration

Test Your Understanding 39

Two balls are projected upwards simultaneously. Ball A starts 50 m above ground at 2 m/s; ball B starts at ground level at 20 m/s. Model both as particles moving freely under gravity. Find the time and height when they are level.

Time

Height

Exercise 9E – Y1 Mech Pages 151-152 40

10. A particle is projected upwards at 24.5 m/s. Find the total time it is at least 21 m above its launch point.

10

11. A particle is projected upwards at u m/s. After 2 s it is still rising at $u/3$ m/s. Find u and the time to return to the launch point.

11

12. Ball A is thrown downwards at 5 m/s from a tower 46 m high. Simultaneously ball B is thrown upwards from the ground at 18 m/s. Find the distance from the top of the tower to their collision point. Neglect air resistance.

Time

Distance

13. A ball is dropped from rest 10 m above a wooden floor. After each impact it rebounds with three-quarters of its impact speed. Find its maximum height after:
    a. the first bounce;
    b. the second bounce. Neglect air resistance.

a

b